

Answer **all** the questions and marks are assigned against each question.

Solve the numerical first on rough paper then write the steps and solution neatly here

1. A 20-foot diameter circular shaft is planned in a massive rock. Laboratory tests on core from exploration drilling show that $C_o = 22,000$ psi; $T_o = 22,000$ psi; $\gamma = 144$ pcf; $E = 5 \times 10^6$ psi; $G = 2.0 \times 10^6$ psi. Depth is 3,000 ft. Site measurements show that no tectonic stresses are present.
- (a) Determine the factors of safety with respect to failure in compression and tension.
- (b) If the opening is in an *insitu* stress field such that $\sigma_h = 0.01\sigma_v$, is there a possibility of failure? Justify your answer. (2 + 2 + 1 = 5 marks)

$$(a) FS = \frac{\text{Strength}}{\text{Stress}}$$

$$FS_c = \frac{C_o}{\sigma_c} \text{ and } FS_t = \frac{T_o}{\sigma_t}$$

$$\sigma_c = K_c S_1 \text{ and } \sigma_t = K_t S_1$$

$$S_v = \gamma H = 3000 \text{ psi}$$

$$S_H = S_h = K_o S_v = 3000 K_o$$

(1) In vertical section, neglecting shaft bottom effect:

$$FS_c = \frac{C_o}{\sigma_c} = \frac{22000}{3000} = 7.33$$

$$E = \text{Young's modulus} = 5.0 \times 10^6 \text{ psi;}$$

$$G = \text{Shear Modulus} = 2.0 \times 10^6 \text{ psi}$$

$$\nu = \text{Poisson's ratio}$$

In plan view:

$$E_{\epsilon_h} = 0 = S_h - \nu S_H - \nu S_v = (1 - \nu) S_h - \nu S_v$$

$$S_h = S_H = \left(\frac{\nu}{1 - \nu} \right) S_v$$

$$G = \frac{E}{2(1 + \nu)} = \frac{5 \times 10^6}{2(1 + \nu)}$$

$$2.0 \times 10^6 = \frac{5 \times 10^6}{2(1 + \nu)}$$

$$\therefore \nu = \frac{1}{4} = 0.25$$

$$S_h = S_H = \left(\frac{0.25}{1 - 0.25} \right) S_v = \frac{1}{3} S_v = 1000 \text{ psi}$$

$$\sigma_{\theta\theta} = \left(\frac{S_h + S_H}{2} \right) (2) - \left(\frac{S_h - S_H}{2} \right) (4) \cos(2\theta) = 2S_h = 2K_o S_v = (2) \left(\frac{1}{3} \right) \gamma H = (2) \left(\frac{1}{3} \right) \gamma H$$

$$FS_c = \frac{C_o}{\sigma_c} = \frac{22000}{2 \times 1000} = 11$$

No tension

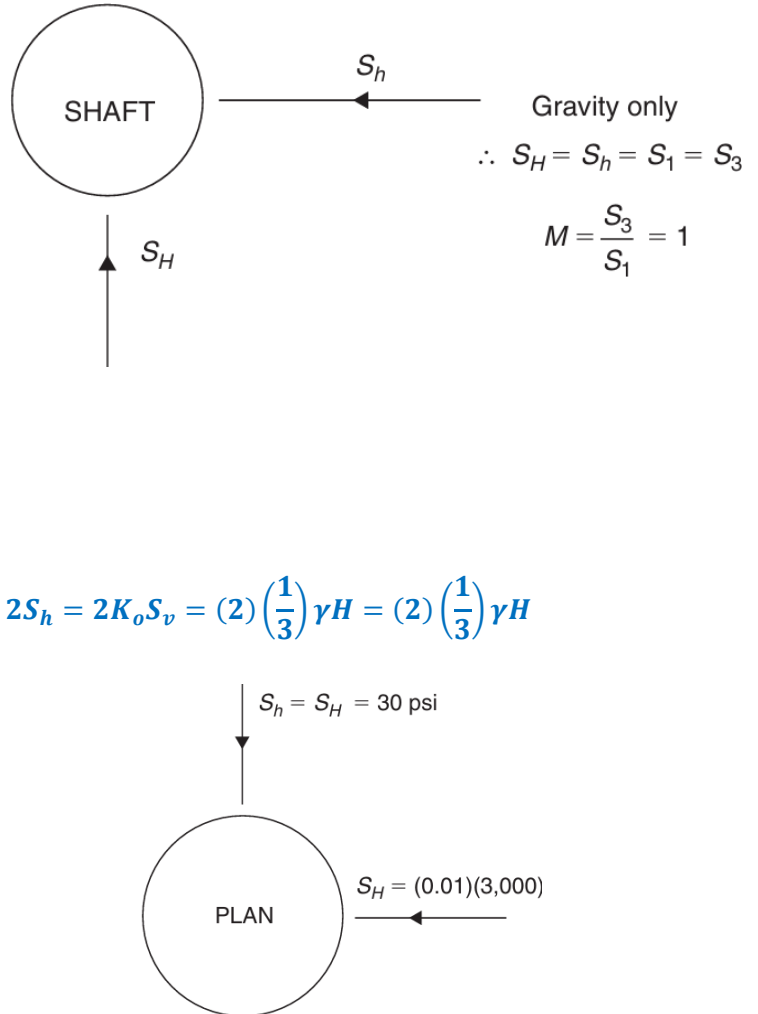
FS_t is not a factor.

$$(b) S_H = S_h = 0.01 S_v$$

$$M = 1; K_c = 2$$

$$FS_c = \frac{C_o}{\sigma_c} = \frac{22000}{2 \times 30} = 361$$

FS_t is not a factor.



2. Coal pillar strength is represented by $S = S_1 h^\alpha w^\beta$, where S_1 = insitu strength of the pillar, h = mining height, and w = pillar width. Two bord and pillar panels are developed in the similar geological conditions at depths D_1 and D_2 with mining heights h_1 and h_2 respectively. If the gallery width and the pillar width in both the panels remain the same, the ratio of pillar safety factors, SF_1/SF_2

(4 marks)

Since gallery width and pillar width for both the panels remain the same, therefore, recovery (R) will also remain the same.

For panel-1

Strength of pillar, $S = S_1 h_1^\alpha w^\beta$

Load acting on pillar = $\frac{\gamma D_1}{1 - R}$

$$SF_1 = \frac{S_1 h_1^\alpha w^\beta}{\frac{\gamma D_1}{1 - R}}$$

For panel-2

Strength of pillar, $S = S_1 h_2^\alpha w^\beta$

Load acting on pillar = $\frac{\gamma D_2}{1 - R}$

$$SF_2 = \frac{S_1 h_2^\alpha w^\beta}{\frac{\gamma D_2}{1 - R}}$$

$$\frac{SF_1}{SF_2} = \frac{\frac{S_1 h_1^\alpha w^\beta}{\frac{\gamma D_1}{1 - R}}}{\frac{S_1 h_2^\alpha w^\beta}{\frac{\gamma D_2}{1 - R}}} = \left(\frac{h_1}{h_2}\right)^\alpha \frac{D_2}{D_1}$$

3. A room and pillar coal mine at a depth of 1,560 ft (475 m) develops pillars 30 ft (9.14 m) wide and 60 ft (18.3 m) long in a flat seam 12 ft (3.66 m) thick. Entries and crosscuts are 24 ft (7.32 m) wide. Laboratory tests of trona core 2-1/8 in. (5.4 cm) in diameter and 4-1/4 in. (10.8 cm) long give an average unconfined compressive strength of 6,740 psi (46.5 MPa) and a coefficient of variation of 38%. A size effect is indicated. If a pillar safety factor with respect to compressive failure of 1.67 is required, estimate the maximum allowable extraction ratio? Use $C_p = C_1 \left(0.78 + 0.22 \frac{W_p}{H_p}\right)$.

(5 marks)

$$S_v = 1 * 1560 = 1560 \text{ psi}$$

$$W_p = \frac{4A}{P} = \frac{4 * 30 * 60}{2(30 + 60)} = 40 \text{ ft}$$

$$C_p = C_1 \left(0.78 + 0.22 \frac{W_p}{H_p}\right) = 6740 \left(0.78 + 0.22 \frac{40}{12}\right) = 10199.87 \text{ psi}$$

$$FS_c = 1.67 = \frac{C_p}{S_p} = \frac{C_p}{\frac{S_v}{1 - R}} = \frac{C_p(1 - R)}{S_v} = \frac{10199.87(1 - R)}{1560} = 6.54(1 - R)$$

$$R = 0.75 = 75\%$$

4. Design a Bord & Pillar panel (number of pillars along length and width) for assumed pillar size of 25 m x 25 m developed in a 4 m thick coal seam with incubation period of 6 months to be extracted using Continuous Miner Technology in Bord and Pillar Mining Method. Assume pillar recovery during depillaring as 70%, working days of 300 in a year and daily desired production of 2000 tons from a Continuous Miner panel. Use specific gravity of coal = 1.3 t/m³.

(5 marks)

Incubation period of the seam= 6 months

Output per day from the panel= 2000 tons

Number of working days = $\frac{300}{12} = 25$ days per month

Total number of working days in 6 months = 6 x 25 = 150 days

Total coal obtained from the panel in 6 months = 1500 x 2000 = 3,00,000 tons

Recovery from the panel = 70%

Total coal reserve insitu (in proposed panel) = 3,00,000 x $\frac{100}{70}$ t = 4,28,571 tons

Coal in a pillar = $w_1 \times w_2 \times h \times g = 25 \text{ m} \times 25 \text{ m} \times 4 \text{ m} \times 1.3 \text{ kg/m}^3 = 3,250 \text{ t}$

Number of pillars in a panel = $\frac{\text{Total coal in-situ}}{\text{Coal in a pillar}} = \frac{428571}{3250} = 131.868 \cong 132$ pillars

4 pillars along width and 33 pillars along length

5. A rectangular shaft 12 ft by 24 ft is planned for deepening from a depth of 3,200 to 4,800 ft in ground where the preshaft stress (in psi) state relative to compass coordinates (x = east, y = north, z = up) is given by

$$S_{xx} = 2,000 + 1.1d; S_{yy} = 50 + 0.9d; S_{zz} = 1.15d; T_{xy} = -350 - 0.3d; T_{yz} = 0; T_{zx} = 0$$

where compression is positive and d is depth (in ft) below surface.

(a) Find the best orientation of the shaft at a planned depth of 3,850 ft and show by sketch.

(b) Also show on the sketch where the peak stress concentrations (both compressive and tensile) are likely to occur.

(c) Finally, estimate the peak stress concentrations. (Use $K_c = 3.29$ and $K_t = 0.18$ for the given conditions)

+ 1 + 2 = 5 marks)

(a) At 3850 ft

$$S_{xx} = 2,000 + 1.1d = 6235 \text{ psi}$$

$$S_{yy} = 50 + 0.9d = 3515 \text{ psi}$$

$$S_{zz} = 1.15d = 4427.5 \text{ psi}$$

$$T_{xy} = -350 - 0.3d = -1505 \text{ psi}$$

$$T_{yz} = 0; T_{zx} = 0$$

$$S_{1,3} = \frac{S_{xx} + S_{yy}}{2} \pm \left[\left(\frac{S_{xx} - S_{yy}}{2} \right)^2 + (T_{xy})^2 \right]^{1/2}$$

$$S_1 = 4875 \pm 2028.45 = 6903.45 \text{ or } 2846.55$$

$$\tan(2\beta) = \left[\frac{T_{xy}}{(S_{xx} - S_{yy})/2} \right]$$

$$= \left[\frac{1505}{(6235 - 3515)/2} \right]$$

$$2\beta = 47.89^\circ$$

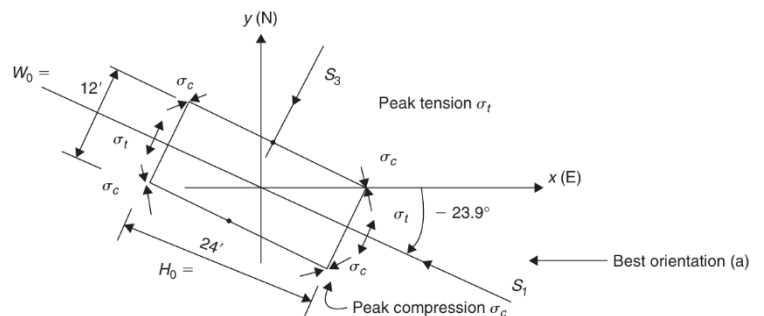
$$\therefore \beta = 23.94^\circ$$

$$(c) M = \frac{S_3}{S_1} = \frac{2847}{6904} = 0.412$$

$$\text{Aspect Ratio} = \frac{W_o}{H_o} = 0.5$$

$$\sigma_c = K_c S_1 = 3.29 * 6904 = 22714 \text{ psi}$$

$$\sigma_t = K_t S_1 = 0.18 * 6904 = +1234 \text{ psi (tension contradicting -since sign is positive and we have considered +ve as compression)}$$



6. Suppose that the average vertical pillar stress is 26.7 MPa in a room and pillar mine and that a joint set is present with an average bearing of N45°W. If the joints obey a Mohr–Coulomb slip criterion and have a friction angle of 28° and a cohesion of 0.90 MPa, what range of joint dips is unsafe with respect to joint slip? (4 marks)

$$\tau_j(\text{stress}) = \left(\frac{S_p}{2}\right) \sin(2\delta)$$

$$\sigma_j(\text{stress}) = \left(\frac{S_p}{2}\right) \left(\frac{1 + \cos(2\delta)}{2}\right)$$

$$\tau_j(\text{strength}) = \sigma_j \tan \phi_j + c_j = \left(\frac{S_p}{2}\right) \left(\frac{1 + \cos(2\delta)}{2}\right) \tan \phi_j + c_j$$

$$\tau_j(\text{stress}) = \left(\frac{S_p}{2}\right) \sin(2\delta) > \tau_j(\text{strength}) = \left(\frac{S_p}{2}\right) \left(\frac{1 + \cos(2\delta)}{2}\right) \tan \phi_j + c_j$$

$$\left(\frac{S_p}{2}\right) \sin(2\delta) > \left(\frac{S_p}{2}\right) \left(\frac{1 + \cos(2\delta)}{2}\right) \tan \phi_j + c_j$$

where absolute shear stress value is used. After rearrangement, this criterion is

$$\sin(2\delta - \phi_j) > \sin(\phi_j) + \frac{2c_j}{S_p} \cos(\phi_j) = \sin(28) + \frac{2 * 0.90}{26.7} \cos(28) = 0.528$$

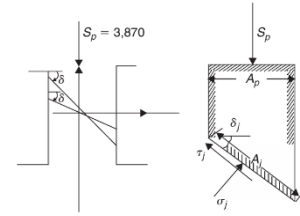
$$2\delta - \phi_j > 31.94$$

$$\delta > 29.96^\circ \dots 1\text{st solution}$$

$$180 - (2\delta - \phi_j) > 31.94$$

$$\delta < 88^\circ \dots 2\text{nd solution}$$

$$\therefore \text{unsafe for } 30^\circ \leq \delta \leq 88^\circ$$



7. Design support system by mentioning and showing grid pattern and number of bolts for galleries and junctions developed using continuous miner technology with the help of following information provided:

Rock Mass Rating = 45; Gallery width = 6.0 m; Rock density, $\gamma = 2500 \text{ kg/m}^3$; Grouting strength of resin material = 20 tons; Factor of Safety = 2

Following relationship may be useful:

$$\text{Rock load in roadways} = B\gamma(1.7 - 0.037RMR + 0.0002RMR^2)$$

$$\text{Rock load in junctions} = 1.25 * \text{Rock load in roadways}$$

(4 marks)

$$\text{Rock load in roadways} = B\gamma(1.7 - 0.037RMR + 0.0002RMR^2) = 6 * 2.5 * (1.7 - 0.037 * 45 + 0.0002 * 45^2) = 6.6 \text{ tons/m}^2$$

Let number of bolts = n; grouting strength is 20 tons for each bolt and grid pattern as 1.2 m x 1.2 m

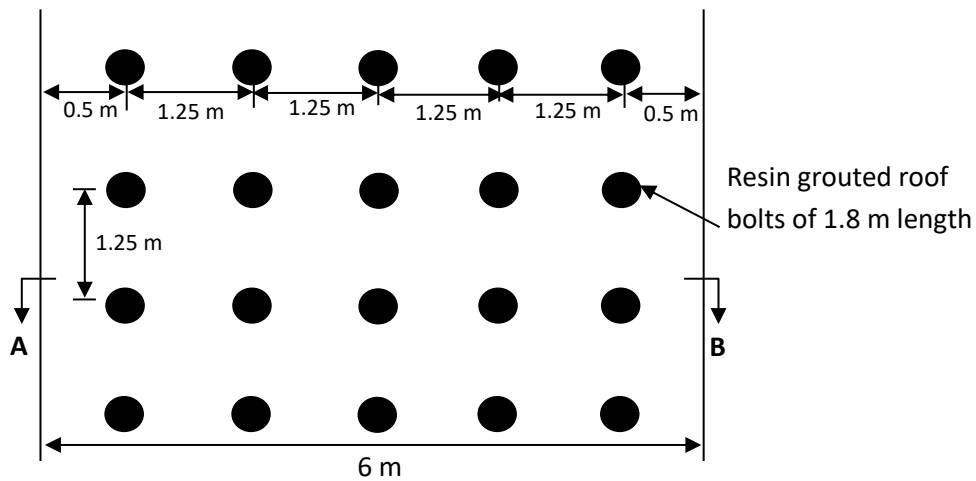
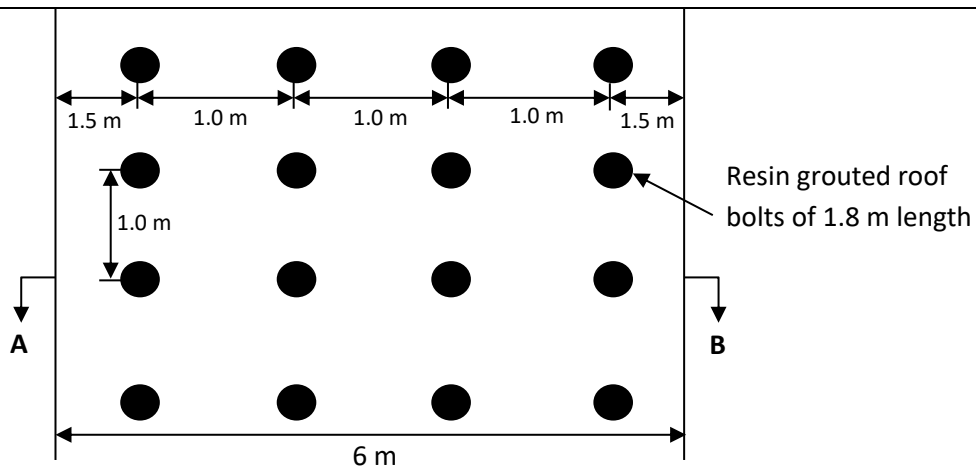
$$\text{Support Resistance } (S_r) = \frac{\text{Number of support} \times \text{Capacity of each support}}{\text{Area supported by one row of supports}} = \frac{n * 20}{6 * g}$$

$$\text{Support safety factor} = \frac{\text{Support Resistance}}{\text{Rock Load}} = \frac{\frac{n * 20}{6 * g}}{6.6} = 2$$

$$n = \frac{6.6 * 2 * 6 * g}{20} = 3.96g$$

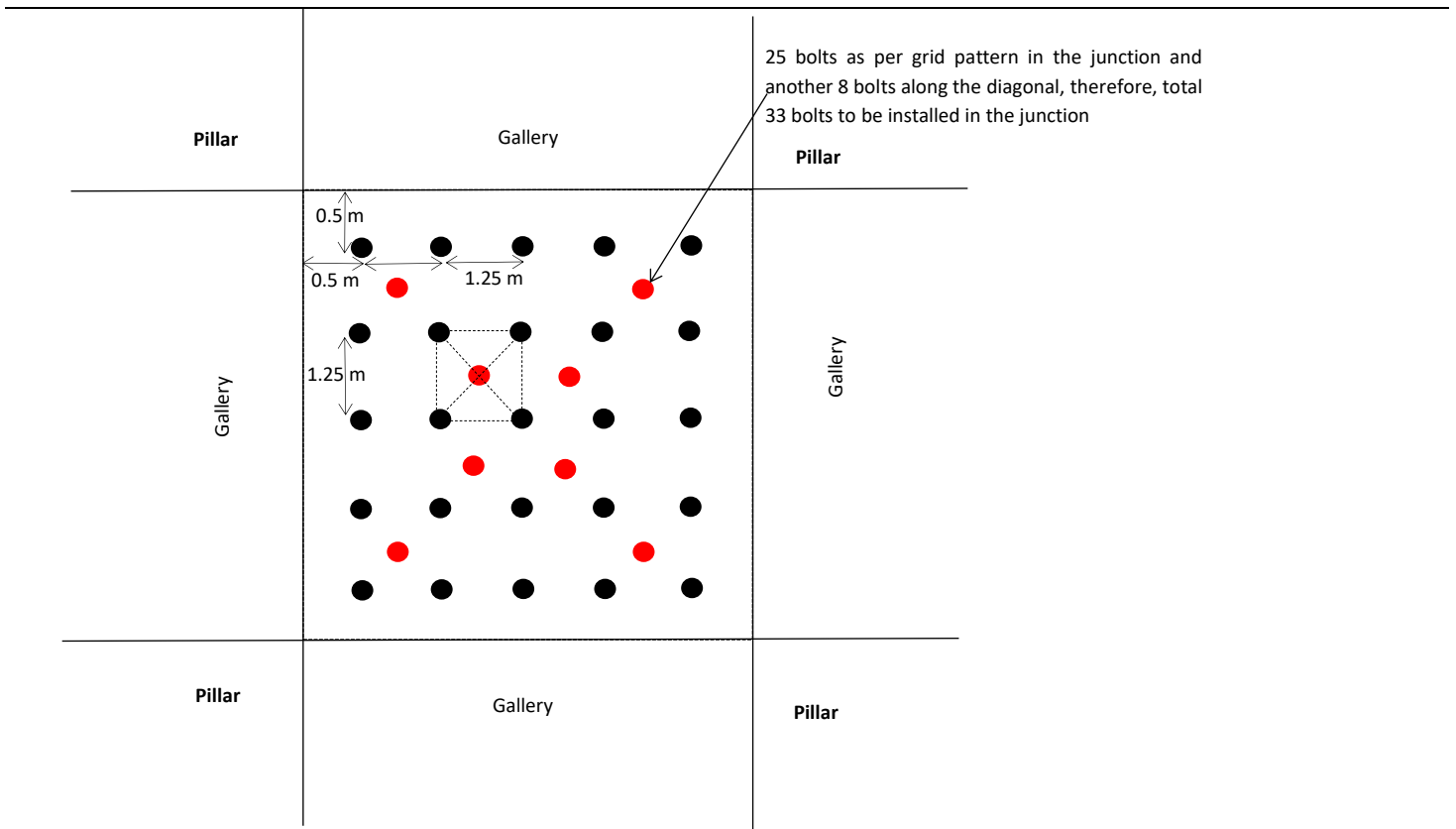
For $g = 1.00$; no. of bolts ≈ 4 not applicable as there is too much spacing between two bolts in a row and between two rows and sides of the pillars

For $g = 1.25$; no. of bolts ≈ 5 , it is applicable



*Rock load in junctions = $1.25 * 6.6 = 8.25 \text{ tons/m}^2$*

$$n = \frac{8.25 * 2 * 6 * 6}{20} = 29.7 \approx 30 \text{ bolts}$$



8. The Cavability Index of a roof was found to be 7000. Does this indicate a problematic roof? If so, discuss the measures to address such a roof.

(1 + 1 = 2 marks)

The Cavability Index (CI) is a measure used to assess the stability and potential for caving in underground mining operations, particularly in relation to the roof of a mine. A CI value of 7000 is considered high and typically indicates a roof caveable with difficulty and weighting may or may not be dynamic in nature and may cause side spalling of pillars. It may require induced caving by roof blasting even after optimum design of rib/snook and breaker-line support.

9. Define side spalling of pillar. Discuss the factors responsible for its occurrence. How it affects the design of rib/snook and breaker-line support?

(1 + 1 + 1 = 3 marks)

SSP, which refers to the fracturing and detachment of material from the sides of the coal pillars, often exacerbated by various geological and operational factors. It typically results from the redistribution of stress around the pillars due to mining activities and is influenced by factors like the geotechnical properties of rock mass, mining depth, pillar shape and size, and roof loading. Additionally, spalling can reduce the effective size of the pillars, compromising their load-bearing capacity and potentially leading to more extensive structural failures within the mine. Side spalling of a pillar refers to the flaking, cracking, or breaking off of material from the sides of a pillar in underground mining. This phenomenon occurs due to the redistribution of stress around the pillar, leading to the failure of the pillar's surface layers. It is a common issue in high-stress environments and can compromise the stability of the pillar and the surrounding mines structure.

Factors Responsible for Side Spalling:

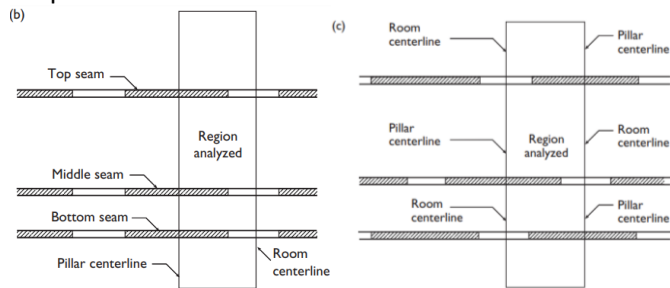
- (a) High Stress Conditions
- (b) Pillar Geometry
- (c) Rock Properties
- (d) Mining Depth
- (e) Time-Dependent Behavior
- (f) Blasting and Vibrations
- (g) Water and Weathering

Impact on the Design of Rib/Snook and Breaker-Line Support:

Side spalling affects the design and stability of rib/snook (the coal or rock left to support the roof) and breaker-line support (the line where the roof is expected to break and cave). The following considerations are important:

- (a) Extent of spalling reduces the effective load-bearing area of pillar and consequently the area of proposed rib/snook
- (b) Shifting of RBBLs towards outby side considering the extent of side spalling of pillars
- (c) Goaf encroachment towards the working during roof falls
- (d) Pillar Width Adjustment
- (e) Stress Redistribution
- (f) Monitoring and Maintenance
- (g) Breaker-Line Stability
- (h) Mining Sequence

10. Discuss staggered and superimposed design of pillars.



(1 mark)

Columnize the mains and stagger the chains. Multi-seam mining involves simultaneous or sequential excavation in two or more seams.

Parting distance in multi-seam workings is essential. The parting should be minimum 9 m (or double of the entry width) for interaction between contiguous working. Stability of main entries and pillars is essential to mine safety and these should be standing intact till life of the mine. With respect to stress concentration "columnizing the mains" is advantageous although if bed separation occurs in the immediate roof then there is lessened advantage and staggered is preferred. In staggered case (b), less separation leads to reduction in width of pillar by the gallery widths.

11. Mention the factors affecting manner of pillar extraction during mechanized pillar extraction in Bord and Pillar Mining Method.

(1 mark)

- (a) Cut-out distance
- (b) Hanging roof span inside the goaf
- (c) Depth
- (d) Rib/snook design
- (e) Breaker-line design
- (f) Pillar strength
- (g) Technology
- (h) Ventilation
- (i) Cost and recovery

12. List the equipment that constitutes the complete package for mass production technology in the Bord and Pillar mining method.	<i>(1 mark)</i> (a) Continuous Miner for cutting – 1 no. (b) Twin/Quad Roof bolter for bolts installation – 1 no. (c) Ram/Shuttle Car – 2 nos. (d) Feeder Breaker – 1 no. (e) Puller to retrieve trapped CM in roof debris – 1 no.
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*****Question Paper Ends Here*****